Full Stack Assignment – 08

Assignment ID: FSD-08

Title: Internal Announcement Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Internal Announcement Management System** that allows administrators to publish announcements and employees to view them.
The system must handle announcement expiry correctly and ensure **data persistence using a database**.

2️⃣ User Roles

EMPLOYEE

- Register and log in to the system
- View active announcements

ADMIN

- Log in to the system
- Create and manage announcements
- Control announcement visibility

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Announcement Management

- ADMIN can create announcements with:
  - Title
  - Description
  - Expiry date
- All announcement data must be stored in the database
- Only active announcements should be visible to EMPLOYEE users

4 Business Rule (Critical)

⚠️ **Expired announcements must not be visible to employees.**

The system must automatically filter announcements based on the expiry date.

5 Technical Constraints (Mandatory)

- All announcement data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect real-time database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6 Frontend Requirements

- Login and registration pages
- Role-based dashboards for EMPLOYEE and ADMIN
- Announcement listing for EMPLOYEE
- Announcement creation and management view for ADMIN

UI appearance is not evaluated; functional clarity is the focus.

7 Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections

- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **date-based logic, role enforcement, and end-to-end correctness**.